Class 12 Mathematics

Chapter 5: Continuity and Differentiability

Practice Set B: step-wise solutions

20 questions **51** marks **3** levels

Each row is one line of working. The red number in the right margin is what an examiner gives for that line, as in the CBSE marking scheme (½ is half a mark). Write every line that carries a mark. ✓ ½

Level 1

8 marks

1 **MCQ** CBSE SQP 2023-24 — 1 mark

If $f(x) = \begin{cases} \dfrac{kx}{|x|}, & \text{if } x < 0 \\ 3, & \text{if } x \geq 0 \end{cases}$ is continuous at $x = 0$, then the value of k is:

(A) -3 **(B)** 0

(C) 3 **(D)** any real number

Formula / idea: Continuous at 0 means LHL $=$ RHL $= f(0)$.

1 For $x < 0$, $|x| = -x$.
$$\frac{kx}{|x|} = \frac{kx}{-x} = -k$$

2 Find the LHL, the RHL and $f(0)$.
$$\lim_{x \to 0^-} f(x) = -k, \qquad \lim_{x \to 0^+} f(x) = 3 = f(0)$$

3 Equate LHL to $f(0)$. ✓ 1
$$-k = 3 \;\Rightarrow\; k = -3$$

4 So the correct option is (A).

Answer **(A)** -3

Watch out: Taking $|x| = x$ for $x < 0$ gives $k = 3$ (option C).

2 MCQ NCERT Exemplar Ex 5.3 Q90 — 1 mark

Let $f(x) = |\sin x|$. Then

(A) f is everywhere differentiable

(B) **f is everywhere continuous but not differentiable at $x = n\pi$, $n \in \mathbb{Z}$**

(C) f is everywhere continuous but not differentiable at $x = (2n+1)\dfrac{\pi}{2}$, $n \in \mathbb{Z}$

(D) none of these

Formula / idea: $|u|$ can have a corner only where $u = 0$; check LHD and RHD there.

1. f is $|x|$ composed with $\sin x$, both continuous, so f is continuous everywhere.
   A composite of continuous functions is continuous
2. A corner can occur only where $\sin x = 0$, i.e. at $x = n\pi$.
3. At $x = n\pi$: $f(n\pi + h) = |\sin h|$ and $f(n\pi) = 0$.
   $\text{LHD} = \lim\limits_{h\to 0} \dfrac{|\sin(-h)|}{-h} = -1, \qquad \text{RHD} = \lim\limits_{h\to 0} \dfrac{|\sin h|}{h} = 1$
4. LHD $\neq$ RHD, so f is not differentiable at $x = n\pi$. ✓ 1
5. So the correct option is (B).

Answer **(B) f is everywhere continuous but not differentiable at $x = n\pi$, $n \in \mathbb{Z}$**

Watch out: Choosing (C): at $x = (2n+1)\frac{\pi}{2}$, $\sin x = \pm 1$, so $|\sin x|$ is smooth there.

3 MCQ Board-pattern — 1 mark

The derivative of $2^{\cos^2 x}$ with respect to x is:

(A) $2^{\cos^2 x} \sin 2x \log 2$

(B) $-\cos^2 x \cdot 2^{\cos^2 x - 1} \sin 2x$

(C) $-2^{\cos^2 x} \sin 2x \log 2$

(D) $-\dfrac{2^{\cos^2 x} \sin 2x}{\log 2}$

Formula / idea: $\dfrac{d}{dx} a^u = a^u \log a \cdot \dfrac{du}{dx}$.

1. Use the formula for a^u with $u = \cos^2 x$.
   $\dfrac{d}{dx} 2^{\cos^2 x} = 2^{\cos^2 x} \log 2 \cdot \dfrac{d}{dx}(\cos^2 x)$
2. Differentiate $\cos^2 x$ (chain rule).
   $\dfrac{d}{dx} \cos^2 x = 2\cos x(-\sin x) = -\sin 2x$

3 Combine. ✓ 1

$$\frac{d}{dx}2^{\cos^2 x} = -2^{\cos^2 x}\ \sin 2x\ \log 2$$

4 So the correct option is (C).

Answer **(C)** $-2^{\cos^2 x}\ \sin 2x\ \log 2$

Watch out: Using the power rule on 2^u gives option (B); the variable is in the exponent.

4 **MCQ** NCERT Exemplar Ex 5.3 Q91 1 mark

If $y = \log\left(\frac{1-x^2}{1+x^2}\right)$, then $\frac{dy}{dx}$ is equal to

(A) $\frac{4x^3}{1-x^4}$ **(B)** $\frac{-4x}{1-x^4}$ **(C)** $\frac{1}{4-x^4}$ **(D)** $\frac{-4x^3}{1-x^4}$

Formula / idea: $\log\frac{m}{n} = \log m - \log n$; then $\frac{d}{dx}\log u = \frac{1}{u}\frac{du}{dx}$.

1 Split the log.

$$y = \log(1-x^2) - \log(1+x^2)$$

2 Differentiate each term (chain rule).

$$\frac{dy}{dx} = \frac{-2x}{1-x^2} - \frac{2x}{1+x^2}$$

3 Take the common denominator $(1-x^2)(1+x^2) = 1-x^4$. ✓ 1

$$\frac{dy}{dx} = \frac{-2x(1+x^2) - 2x(1-x^2)}{1-x^4} = \frac{-4x}{1-x^4}$$

4 So the correct option is (B).

Answer **(B)** $\frac{-4x}{1-x^4}$

Watch out: Writing $\frac{d}{dx}\log(1-x^2) = \frac{2x}{1-x^2}$; the inner derivative is $-2x$, not $2x$.

5 MCQ NCERT Exemplar Ex 5.3 Q94 1 mark

If $x = t^2$, $y = t^3$, then $\dfrac{d^2y}{dx^2}$ is

(A) $\dfrac{3}{2}$ **(B)** $\dfrac{3}{4t}$ **(C)** $\dfrac{3}{2t}$ **(D)** $\dfrac{3}{4}$

Formula / idea: $\dfrac{d^2y}{dx^2} = \dfrac{d}{dt}\left(\dfrac{dy}{dx}\right) \cdot \dfrac{dt}{dx}$.

1 $\dfrac{dx}{dt} = 2t$, $\dfrac{dy}{dt} = 3t^2$. Divide.
$\dfrac{dy}{dx} = \dfrac{3t^2}{2t} = \dfrac{3t}{2}$

2 Differentiate again w.r.t. x (chain rule through t). ✓ 1
$\dfrac{d^2y}{dx^2} = \dfrac{3}{2} \cdot \dfrac{dt}{dx} = \dfrac{3}{2} \cdot \dfrac{1}{2t} = \dfrac{3}{4t}$

3 So the correct option is (B).

Answer **(B)** $\dfrac{3}{4t}$

Watch out: Stopping at $\dfrac{d}{dt}\left(\dfrac{3t}{2}\right) = \dfrac{3}{2}$ (option A) without multiplying by $\dfrac{dt}{dx}$.

6 Assertion-Reason Board-pattern 1 mark

Assertion (A): The function $f(x) = \begin{cases} \dfrac{x}{|x| + 2x^2}, & x \neq 0 \\ k, & x = 0 \end{cases}$ is discontinuous at $x = 0$, whatever the value of k. **Reason (R):** $\lim\limits_{x \to 0^-} f(x) \neq \lim\limits_{x \to 0^+} f(x)$.

(A) **Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of the Assertion (A).**

(B) Both Assertion (A) and Reason (R) are true, but Reason (R) is **not** the correct explanation of the Assertion (A).

(C) Assertion (A) is true, but Reason (R) is false.

(D) Assertion (A) is false, but Reason (R) is true.

Formula / idea: If LHL $\neq$ RHL, the limit does not exist, so no value $f(0)$ can make f continuous.

1 LHL: for $x < 0$, $|x| = -x$.
$\lim\limits_{x \to 0^-} \dfrac{x}{-x + 2x^2} = \lim\limits_{x \to 0^-} \dfrac{1}{-1 + 2x} = -1$

2 RHL: for $x > 0$, $|x| = x$.
$\lim\limits_{x \to 0^+} \dfrac{x}{x + 2x^2} = \lim\limits_{x \to 0^+} \dfrac{1}{1 + 2x} = 1$

3	$-1 \neq 1$, so **R is true**.	
4	The limit at 0 does not exist, so no k works: **A is true**.	
5	A follows directly from R, so **R explains A**.	✓ 1
6	So the correct option is (A).	

Answer **(A) Both A and R are true and R is the correct explanation of A**

Watch out: Cancelling x as if $|x| = x$ on both sides, which gives limit 1 and $k = 1$.

7 **Short answer** NCERT Exemplar Ex 5.3 Q43 2 marks

Differentiate $\tan^{-1}\left(\dfrac{\sqrt{1+x^2}+\sqrt{1-x^2}}{\sqrt{1+x^2}-\sqrt{1-x^2}}\right)$ with respect to x, where $-1 < x < 1, x \neq 0$.

Formula / idea: Put $x^2 = \cos 2\theta$ so the roots become $\sqrt{2}\cos\theta$ and $\sqrt{2}\sin\theta$.

1	Let y be the function. Put $x^2 = \cos 2\theta$, $0 < \theta < \dfrac{\pi}{4}$. $\sqrt{1+\cos 2\theta} = \sqrt{2}\cos\theta, \qquad \sqrt{1-\cos 2\theta} = \sqrt{2}\sin\theta$ $0 < x^2 < 1$ gives $0 < 2\theta < \frac{\pi}{2}$	✓ ½
2	Divide numerator and denominator by $\cos\theta$. $\dfrac{\cos\theta+\sin\theta}{\cos\theta-\sin\theta} = \dfrac{1+\tan\theta}{1-\tan\theta} = \tan\left(\dfrac{\pi}{4}+\theta\right)$	✓ ½
3	So $y = \tan^{-1}\tan\left(\dfrac{\pi}{4}+\theta\right)$; write θ in terms of x. $y = \dfrac{\pi}{4}+\theta = \dfrac{\pi}{4}+\dfrac{1}{2}\cos^{-1}(x^2)$ $\frac{\pi}{4}+\theta$ lies in $\left(\frac{\pi}{4}, \frac{\pi}{2}\right)$, inside the range of $\tan^{-1}$	✓ ½
4	Differentiate w.r.t. x (chain rule). $\dfrac{dy}{dx} = \dfrac{1}{2}\cdot\dfrac{-1}{\sqrt{1-x^4}}\cdot 2x = -\dfrac{x}{\sqrt{1-x^4}}$	✓ ½

Answer $\dfrac{dy}{dx} = -\dfrac{x}{\sqrt{1-x^4}}$

Watch out: Forgetting the factor $2x$ from the inner function x^2 in $\cos^{-1}(x^2)$.

Level 2

18 marks

8 **Short answer** NCERT Exemplar Ex 5.3 Q12 — 2 marks

Find the value of k so that the function $f(x) = \begin{cases} \dfrac{2^{x+2} - 16}{4^x - 16}, & \text{if } x \neq 2 \\ k, & \text{if } x = 2 \end{cases}$ is continuous at $x = 2$.

Formula / idea: Continuity at 2: $\lim\limits_{x \to 2} f(x) = f(2)$; put $2^x = t$ to remove the $\dfrac{0}{0}$ form.

1 Put $2^x = t$; as $x \to 2$, $t \to 4$. ✓ ½

$$\frac{2^{x+2} - 16}{4^x - 16} = \frac{4t - 16}{t^2 - 16}$$

$2^{x+2} = 4 \cdot 2^x$ and $4^x = (2^x)^2$

2 Factorise and cancel $(t - 4)$, since $t \neq 4$. ✓ ½

$$\frac{4(t-4)}{(t-4)(t+4)} = \frac{4}{t+4}$$

3 Take the limit. ✓ ½

$$\lim_{x \to 2} f(x) = \lim_{t \to 4} \frac{4}{t+4} = \frac{4}{8} = \frac{1}{2}$$

4 Equate to $f(2) = k$. ✓ ½

$$k = \frac{1}{2}$$

Answer $k = \dfrac{1}{2}$

Watch out: Writing $4^x = 4 \cdot 2^x$; in fact $4^x = (2^x)^2 = t^2$.

9 **Short answer** NCERT Misc. Ex. Q11 — 3 marks

Differentiate w.r.t. x: $x^{x^2-3} + (x-3)^{x^2}$, for $x > 3$.

Formula / idea: Split $y = u + v$; take log of each power separately.

1 Let $u = x^{x^2-3}$ and $v = (x-3)^{x^2}$, so $y = u + v$. ✓ ½

$$\frac{dy}{dx} = \frac{du}{dx} + \frac{dv}{dx}$$

log of a sum cannot be split

2 Take log of u and differentiate (product rule).

$$\log u = (x^2 - 3)\log x \;\Rightarrow\; \frac{1}{u}\frac{du}{dx} = 2x \log x + \frac{x^2 - 3}{x}$$

3 Multiply by u. ✓ 1

$$\frac{du}{dx} = x^{x^2-3}\left(2x \log x + \frac{x^2 - 3}{x}\right)$$

4 Take log of v and differentiate (product rule).

$$\log v = x^2 \log(x-3) \;\Rightarrow\; \frac{1}{v}\frac{dv}{dx} = 2x\log(x-3) + \frac{x^2}{x-3}$$

5 Multiply by v. ✓ 1

$$\frac{dv}{dx} = (x-3)^{x^2}\left(2x\log(x-3) + \frac{x^2}{x-3}\right)$$

6 Add the two results. ✓ ½

$$\frac{dy}{dx} = x^{x^2-3}\left(2x\log x + \frac{x^2-3}{x}\right) + (x-3)^{x^2}\left(2x\log(x-3) + \frac{x^2}{x-3}\right)$$

Answer $\dfrac{dy}{dx} = x^{x^2-3}\left(2x\log x + \dfrac{x^2-3}{x}\right) + (x-3)^{x^2}\left(2x\log(x-3) + \dfrac{x^2}{x-3}\right)$

Watch out: Using the power rule on x^{x^2-3}; when x is in the exponent, take log first.

10 Short answer NCERT Exemplar Ex 5.3 Q51 2 marks

If $x = 3\sin t - \sin 3t$, $y = 3\cos t - \cos 3t$, find $\dfrac{dy}{dx}$ at $t = \dfrac{\pi}{3}$.

Formula / idea: Parametric form: $\dfrac{dy}{dx} = \dfrac{dy/dt}{dx/dt}$.

1 Differentiate x w.r.t. t. ✓ ½

$$\frac{dx}{dt} = 3\cos t - 3\cos 3t$$

2 Differentiate y w.r.t. t. ✓ ½

$$\frac{dy}{dt} = -3\sin t + 3\sin 3t$$

3 Divide. ✓ ½

$$\frac{dy}{dx} = \frac{\sin 3t - \sin t}{\cos t - \cos 3t}$$

4 Put $t = \dfrac{\pi}{3}$: $\sin\pi = 0$, $\cos\pi = -1$. ✓ ½

$$\left(\frac{dy}{dx}\right)_{t=\frac{\pi}{3}} = \frac{0 - \frac{\sqrt{3}}{2}}{\frac{1}{2} + 1} = -\frac{1}{\sqrt{3}}$$

Answer $\left(\dfrac{dy}{dx}\right)_{t=\frac{\pi}{3}} = -\dfrac{1}{\sqrt{3}}$

Watch out: Writing $\dfrac{d}{dt}\sin 3t = \cos 3t$; the chain rule gives $3\cos 3t$.

11 **Short answer** NCERT Exemplar Ex 5.3 Q81 3 marks

If $x = \sin t$ and $y = \sin pt$, prove that $(1 - x^2)\dfrac{d^2y}{dx^2} - x\dfrac{dy}{dx} + p^2y = 0$.

Formula / idea: Find $\dfrac{dy}{dx}$, square to remove $\cos t$ and $\cos pt$, then differentiate again.

1 Differentiate x and y w.r.t. t. ✓ ½

$$\frac{dx}{dt} = \cos t, \qquad \frac{dy}{dt} = p\cos pt$$

2 Divide and cross-multiply. ✓ ½

$$\frac{dy}{dx} = \frac{p\cos pt}{\cos t} \Rightarrow \cos t\,\frac{dy}{dx} = p\cos pt$$

3 Square; use $\cos^2 t = 1 - x^2$ and $\cos^2 pt = 1 - y^2$. ✓ ½

$$(1 - x^2)\left(\frac{dy}{dx}\right)^2 = p^2(1 - y^2)$$

4 Differentiate both sides w.r.t. x. ✓ 1

$$(1 - x^2)\cdot 2\frac{dy}{dx}\cdot\frac{d^2y}{dx^2} - 2x\left(\frac{dy}{dx}\right)^2 = -2p^2y\frac{dy}{dx}$$

5 Divide by $2\dfrac{dy}{dx}$ and rearrange. Hence proved. ✓ ½

$$(1 - x^2)\frac{d^2y}{dx^2} - x\frac{dy}{dx} + p^2y = 0$$

Valid where $\frac{dy}{dx} \neq 0$

Answer $(1 - x^2)\dfrac{d^2y}{dx^2} - x\dfrac{dy}{dx} + p^2y = 0$ **(proved)**

Watch out: Differentiating $\dfrac{p\cos pt}{\cos t}$ w.r.t. t and forgetting $\dfrac{dt}{dx}$; squaring first avoids this.

12 **Short answer** NCERT Misc. Ex. Q14 3 marks

If $x\sqrt{1+y} + y\sqrt{1+x} = 0$, for $-1 < x < 1$, prove that $\dfrac{dy}{dx} = -\dfrac{1}{(1+x)^2}$.

Formula / idea: Square and factorise to get y explicitly, then differentiate.

1 Move one term across and square. ✓ ½

$$x\sqrt{1+y} = -y\sqrt{1+x} \Rightarrow x^2(1+y) = y^2(1+x)$$

2 Rearrange and factorise. ✓ 1

$$(x^2 - y^2) + xy(x - y) = 0 \Rightarrow (x - y)(x + y + xy) = 0$$

3 $x = y$ fails the given equation (except at $x = 0$), so $x + y + xy = 0$. ✓ ½

$$y = -\frac{x}{1+x}$$

$x = y$ gives $2x\sqrt{1+x} = 0$

4 Differentiate (quotient rule). Hence proved. ✓ 1

$$\frac{dy}{dx} = -\frac{(1+x)\cdot 1 - x\cdot 1}{(1+x)^2} = -\frac{1}{(1+x)^2}$$

Answer $\frac{dy}{dx} = -\frac{1}{(1+x)^2}$ **(proved)**

Watch out: Not rejecting the factor $x - y = 0$; it gives $\frac{dy}{dx} = 1$, which is wrong here.

13 **Short answer** NCERT Exemplar Ex 5.3 Q55 2 marks

Find $\frac{dy}{dx}$ when x and y are connected by the relation $\sec(x+y) = xy$.

Formula / idea: Implicit differentiation: differentiate each side w.r.t. x, treating y as a function of x.

1 Differentiate both sides w.r.t. x (chain rule on the left, product rule on the right). ✓ 1

$$\sec(x+y)\tan(x+y)\left(1+\frac{dy}{dx}\right) = y + x\frac{dy}{dx}$$

2 Collect the $\frac{dy}{dx}$ terms. ✓ ½

$$\frac{dy}{dx}\left[\sec(x+y)\tan(x+y) - x\right] = y - \sec(x+y)\tan(x+y)$$

3 Divide. ✓ ½

$$\frac{dy}{dx} = \frac{y - \sec(x+y)\tan(x+y)}{\sec(x+y)\tan(x+y) - x}$$

Answer $\frac{dy}{dx} = \frac{y - \sec(x+y)\tan(x+y)}{\sec(x+y)\tan(x+y) - x}$

Watch out: Missing the factor $\left(1+\frac{dy}{dx}\right)$ from the chain rule on $\sec(x+y)$.

14 **Short answer** PYQ-style 3 marks

Find the values of a, b and c for which the function $f(x) = \begin{cases} \dfrac{\sin(a+1)x + \sin x}{x}, & x < 0 \\ c, & x = 0 \\ \dfrac{\sqrt{x+bx^2} - \sqrt{x}}{bx^{3/2}}, & x > 0 \end{cases}$ is continuous at $x = 0$.

Formula / idea: Continuous at 0: LHL = RHL = $f(0)$; use $\lim\limits_{\theta\to 0}\dfrac{\sin\theta}{\theta} = 1$ and rationalisation.

1 LHL: split the fraction. ✓ 1

$$\lim_{x\to 0^-}\left[\frac{\sin(a+1)x}{(a+1)x}\cdot(a+1) + \frac{\sin x}{x}\right] = (a+1) + 1 = a + 2$$

$\lim_{\theta\to 0}\frac{\sin\theta}{\theta} = 1$

2 RHL: take $\sqrt{x}$ common from the numerator.

$$\frac{\sqrt{x}\left(\sqrt{1+bx} - 1\right)}{bx\sqrt{x}} = \frac{\sqrt{1+bx} - 1}{bx}$$

3 Rationalise and take the limit. ✓ 1

$$\lim_{x\to 0^+}\frac{(1+bx) - 1}{bx\left(\sqrt{1+bx}+1\right)} = \lim_{x\to 0^+}\frac{1}{\sqrt{1+bx}+1} = \frac{1}{2}$$

4 Equate LHL, RHL and $f(0) = c$. ✓ ½

$$a + 2 = \frac{1}{2} = c \;\Rightarrow\; a = -\frac{3}{2}, \quad c = \frac{1}{2}$$

5 The RHL is $\dfrac{1}{2}$ for every b, so b is any non-zero real number. ✓ ½

$b \neq 0$, else f is undefined for $x > 0$

Answer $a = -\dfrac{3}{2}$, $c = \dfrac{1}{2}$, **and** b **is any non-zero real number**

Watch out: Hunting for one value of b; the right-hand limit equals $\frac{1}{2}$ for every $b \neq 0$.

Level 3

25 marks

15 **Short answer** PYQ-style 3 marks

If $y = \dfrac{\sin^{-1}x}{\sqrt{1-x^2}}$, $-1 < x < 1$, show that $(1-x^2)\dfrac{d^2y}{dx^2} - 3x\dfrac{dy}{dx} - y = 0$.

Formula / idea: Clear the root first: $\sqrt{1-x^2}\,y = \sin^{-1}x$, then differentiate twice.

1 Cross-multiply. ✓ ½

$$\sqrt{1-x^2}\;y = \sin^{-1}x$$

2 Differentiate w.r.t. x (product rule); write $y_1 = \dfrac{dy}{dx}$. ✓ 1

$$\sqrt{1-x^2}\,y_1 - \frac{x}{\sqrt{1-x^2}}\,y = \frac{1}{\sqrt{1-x^2}}$$

3 Multiply by $\sqrt{1-x^2}$. ✓ ½

$$(1-x^2)y_1 - xy = 1$$

4 Differentiate again w.r.t. x; write $y_2 = \dfrac{d^2y}{dx^2}$. ✓ ½

$$(1-x^2)y_2 - 2xy_1 - (y + xy_1) = 0$$

5 Collect the terms. Hence proved. ✓ ½

$$(1-x^2)\frac{d^2y}{dx^2} - 3x\frac{dy}{dx} - y = 0$$

Answer $(1-x^2)\dfrac{d^2y}{dx^2} - 3x\dfrac{dy}{dx} - y = 0$ **(proved)**

Watch out: Writing $\dfrac{d}{dx}(xy) = y$ instead of $y + xy_1$, which gives $-2xy_1$ in place of $-3xy_1$.

16 **Long answer** Board-pattern 5 marks

If $ax^2 + 2hxy + by^2 = 1$, prove that $\dfrac{d^2y}{dx^2} = \dfrac{h^2 - ab}{(hx+by)^3}$.

Formula / idea: Find y_1 implicitly, differentiate it by the quotient rule, then use the given equation to simplify.

1 Differentiate both sides w.r.t. x. ✓ 1

$$2ax + 2h\left(y + x\frac{dy}{dx}\right) + 2by\frac{dy}{dx} = 0$$

2 Solve for $\dfrac{dy}{dx}$ (write y_1). ✓ ½

$$y_1 = -\frac{ax+hy}{hx+by} \quad \cdots(1)$$

3 Differentiate (1) w.r.t. x (quotient rule); write $y_2 = \dfrac{d^2y}{dx^2}$. ✓ 1

$$y_2 = -\frac{(a+hy_1)(hx+by) - (ax+hy)(h+by_1)}{(hx+by)^2}$$

4 Expand the numerator N; the ahx and $bhyy_1$ terms cancel. ✓ 1

$$N = ahx + aby + h^2xy_1 + bhyy_1 - ahx - h^2y - abxy_1 - bhyy_1$$
$$= (ab - h^2)(y - xy_1)$$

5 Use (1) and the given equation. ✓ 1

$$y - xy_1 = y + \frac{x(ax+hy)}{hx+by} = \frac{ax^2+2hxy+by^2}{hx+by} = \frac{1}{hx+by}$$

$$ax^2 + 2hxy + by^2 = 1$$

6 Substitute in y_2. Hence proved. ✓ ½

$$\frac{d^2y}{dx^2} = -\frac{ab-h^2}{(hx+by)^2}\cdot\frac{1}{hx+by} = \frac{h^2-ab}{(hx+by)^3}$$

Answer $\dfrac{d^2y}{dx^2} = \dfrac{h^2-ab}{(hx+by)^3}$ **(proved)**

Watch out: Trying to simplify y_2 without using $ax^2+2hxy+by^2=1$; the numerator only collapses after that.

17 **Short answer** NCERT Exemplar Ex 5.3 Q61 3 marks

If $y = (\cos x)^{(\cos x)^{(\cos x)^{\cdots\infty}}}$, show that $\dfrac{dy}{dx} = \dfrac{y^2\tan x}{y\log\cos x - 1}$.

Formula / idea: An infinite tower repeats itself, so the exponent is y: $y = (\cos x)^y$.

1 The exponent is the same infinite tower, i.e. y itself. ✓ ½

$$y = (\cos x)^y$$

Removing one $\cos x$ leaves the same tower

2 Take log on both sides. ✓ ½

$$\log y = y\log\cos x$$

3 Differentiate w.r.t. x (product rule on the right). ✓ 1

$$\frac{1}{y}\frac{dy}{dx} = \frac{dy}{dx}\log\cos x + y\cdot\frac{-\sin x}{\cos x}$$

4 Collect the $\dfrac{dy}{dx}$ terms. ✓ ½

$$\frac{dy}{dx}\cdot\frac{1-y\log\cos x}{y} = -y\tan x$$

5 Solve for $\dfrac{dy}{dx}$. Hence proved. ✓ ½

$$\frac{dy}{dx} = \frac{-y^2\tan x}{1-y\log\cos x} = \frac{y^2\tan x}{y\log\cos x - 1}$$

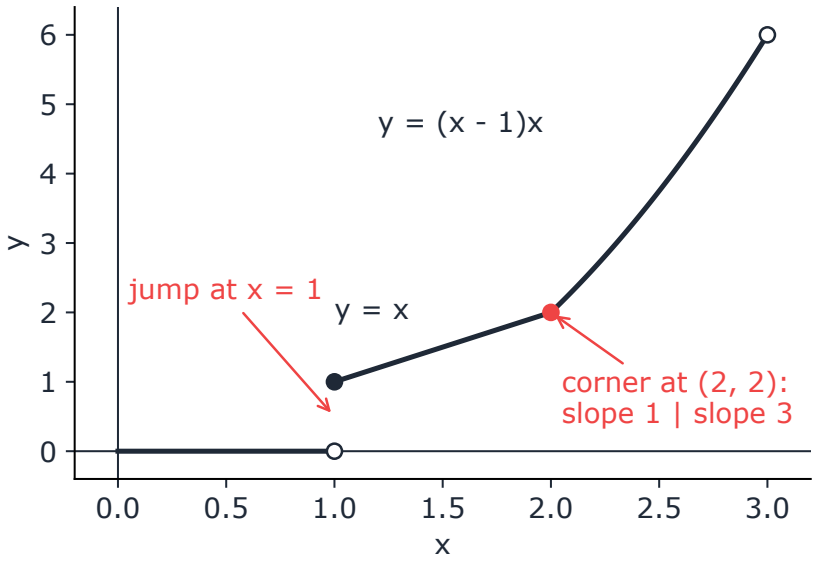
Answer $\dfrac{dy}{dx} = \dfrac{y^2\tan x}{y\log\cos x - 1}$ **(proved)**

Watch out: Treating the exponent y as a constant; $y\log\cos x$ needs the product rule.

18 **Long answer** Board-pattern 5 marks

A function f is defined on $[0,3)$ by $f(x)=\begin{cases}x[x], & 0\le x<2\\ (x-1)x, & 2\le x<3\end{cases}$ where $[x]$ is the greatest integer function. Examine the continuity and differentiability of f at $x=1$ and at $x=2$.

Formula / idea: Remove $[x]$ piece by piece; differentiable needs continuity first and then LHD $=$ RHD.



1 Write f without $[x]$. ✓ 1

$$f(x)=\begin{cases}0, & 0\le x<1\\ x, & 1\le x<2\\ x^2-x, & 2\le x<3\end{cases}$$

$[x]=0$ on $[0,1)$ and $[x]=1$ on $[1,2)$

2 At $x=1$: find the one-sided limits and $f(1)$. ✓ 1

$\lim\limits_{x\to1^-} f(x)=0, \qquad \lim\limits_{x\to1^+} f(x)=1=f(1)$

3 LHL $\neq f(1)$, so f is discontinuous at $x=1$; hence not differentiable there. ✓ ½

Differentiable at a point implies continuous there

4 At $x=2$: find the one-sided limits and $f(2)$. ✓ ½

$\lim\limits_{x\to2^-} x=2, \qquad \lim\limits_{x\to2^+}(x^2-x)=2, \qquad f(2)=2$

5 All three are equal, so f is continuous at $x=2$. ✓ ½

6 LHD at $x=2$ (use $f(x)=x$ on the left). ✓ ½

$$\text{LHD}=\lim_{h\to0}\frac{f(2-h)-f(2)}{-h}=\lim_{h\to0}\frac{(2-h)-2}{-h}=1$$

7 RHD at $x=2$ (use $f(x)=x^2-x$ on the right). ✓ ½

$$\text{RHD}=\lim_{h\to0}\frac{(2+h)^2-(2+h)-2}{h}=\lim_{h\to0}\frac{3h+h^2}{h}=3$$

8	LHD $\neq$ RHD, so f is continuous but not differentiable at $x = 2$.	✓ ½

Answer **At $x = 1$: discontinuous, so not differentiable. At $x = 2$: continuous but not differentiable (LHD $= 1$, RHD $= 3$).**

Watch out: Taking $[x] = 2$ just to the left of $x = 2$; for $1 \le x < 2$, $[x] = 1$.

19 **Long answer** Board-pattern — 5 marks

If $x = a(\cos t + t\sin t)$ and $y = a(\sin t - t\cos t)$, $0 < t < \dfrac{\pi}{2}$, find $\dfrac{d^2y}{dx^2}$. Hence find its value at $t = \dfrac{\pi}{4}$.

Formula / idea: $\dfrac{d^2y}{dx^2} = \dfrac{d}{dt}\left(\dfrac{dy}{dx}\right)\cdot\dfrac{dt}{dx}$.

1	Differentiate x w.r.t. t (product rule on $t\sin t$). $\dfrac{dx}{dt} = a(-\sin t + \sin t + t\cos t) = at\cos t$	✓ 1
2	Differentiate y w.r.t. t (product rule on $t\cos t$). $\dfrac{dy}{dt} = a(\cos t - \cos t + t\sin t) = at\sin t$	✓ 1
3	Divide. $\dfrac{dy}{dx} = \dfrac{at\sin t}{at\cos t} = \tan t$	✓ ½
4	Differentiate again w.r.t. x (chain rule through t). $\dfrac{d^2y}{dx^2} = \sec^2 t\cdot\dfrac{dt}{dx} = \sec^2 t\cdot\dfrac{1}{at\cos t}$	✓ 1
5	Simplify. $\dfrac{d^2y}{dx^2} = \dfrac{\sec^3 t}{at}$	✓ ½
6	Put $t = \dfrac{\pi}{4}$, where $\sec^3\dfrac{\pi}{4} = (\sqrt{2})^3 = 2\sqrt{2}$. $\left(\dfrac{d^2y}{dx^2}\right)_{t=\frac{\pi}{4}} = \dfrac{2\sqrt{2}}{a\cdot\frac{\pi}{4}} = \dfrac{8\sqrt{2}}{a\pi}$	✓ 1

Answer $\dfrac{d^2y}{dx^2} = \dfrac{\sec^3 t}{at}$**; at $t = \dfrac{\pi}{4}$ it is** $\dfrac{8\sqrt{2}}{a\pi}$

Watch out: Writing $\dfrac{d}{dt}(t\sin t) = t\cos t$ and dropping $\sin t$; then the terms do not cancel.

20 **Case study** Board-pattern 4 marks

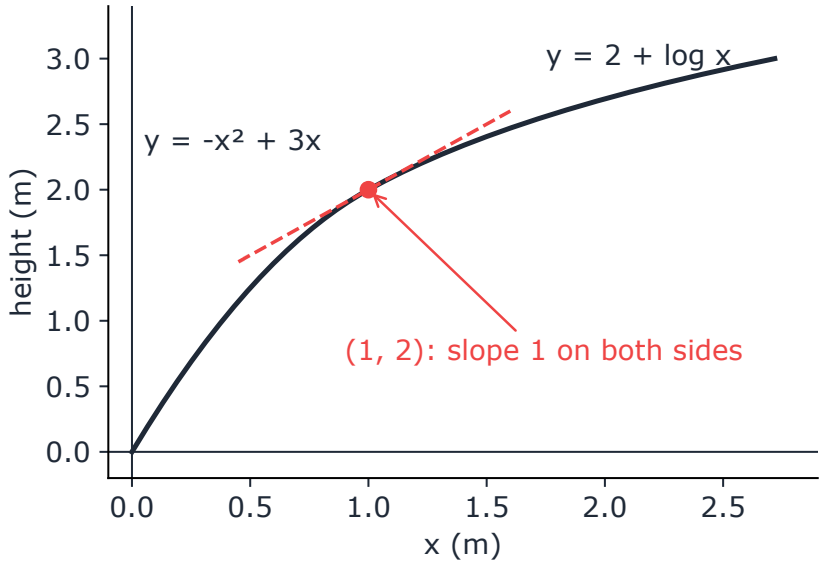
Case Study: A skate-park designer models the side profile of a ramp. If x is the horizontal distance (in m) and $f(x)$ is the height (in m), the profile is $f(x) = \begin{cases} ax^2 + bx, & 0 \le x \le 1 \\ 2 + \log x, & 1 < x \le e \end{cases}$ For a smooth ride the two pieces must meet without a gap or a sharp corner, so f must be continuous and differentiable at $x = 1$.

(i) Write the condition on a and b for f to be continuous at $x = 1$. 1 mark

(ii) Using (i), find the left-hand derivative and the right-hand derivative of f at $x = 1$. 1 mark

(iii) Find the values of a and b, and the height of the ramp at $x = \dfrac{1}{2}$. 2 marks

Formula / idea: Continuity: LHL $=$ RHL $= f(1)$. Differentiability: LHD $=$ RHD. Use $\lim\limits_{h\to 0} \dfrac{\log(1+h)}{h} = 1$.



(i) 1 mark

1 $f(1)$ and the LHL come from the first piece.
$\lim\limits_{x\to 1^-} f(x) = f(1) = a + b$ ✓ ½

2 RHL from the second piece; equate.
$\lim\limits_{x\to 1^+} (2 + \log x) = 2 + 0 = 2 \;\Rightarrow\; a + b = 2$ ✓ ½

Answer (i) $a + b = 2$

(ii) 1 mark

1 LHD, with $f(1) = a + b$.
$\text{LHD} = \lim\limits_{h\to 0} \dfrac{a(1-h)^2 + b(1-h) - (a+b)}{-h} = \lim\limits_{h\to 0}(2a + b - ah) = 2a + b$ ✓ ½

2 RHD, with $f(1) = 2$ from (i). ✓ ½

$$\text{RHD} = \lim_{h \to 0} \frac{2 + \log(1+h) - 2}{h} = \lim_{h \to 0} \frac{\log(1+h)}{h} = 1$$

Answer (ii) **LHD** $= 2a + b$**, RHD** $= 1$

(iii)

2 marks

1 Differentiable at $x = 1$: LHD $=$ RHD. ✓ ½

$$2a + b = 1$$

2 Subtract $a + b = 2$ from it. ✓ 1

$$a = 1 - 2 = -1, \qquad b = 2 - a = 3$$

3 Height at $x = \dfrac{1}{2}$, from $f(x) = -x^2 + 3x$. ✓ ½

$$f\left(\frac{1}{2}\right) = -\frac{1}{4} + \frac{3}{2} = \frac{5}{4}\ \text{m}$$

Answer (iii) $a = -1, b = 3$**; height at** $x = \dfrac{1}{2}$ **is** $\dfrac{5}{4}$ **m**

Watch out: Finding the RHD before using $f(1) = 2$; without continuity the right-hand quotient has no finite limit.